



**Verified Carbon
Standard**

RED SEA POWER WIND FARM

Document prepared by AERA Group

On behalf of Red Sea Power Limited SAS

Project Title	Red Sea Power Wind Farm
Version	V1
Date of Issue	06-May-2022
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1 PROJECT DETAILS

1.1 Summary Description of the Project

The project consists of the construction and operation of a greenfield 58.9MW wind power plant near the Lake of Ghoubet, Djibouti. It involves the setting up of wind turbines which will produce electricity for export to the national grid through EDD (Electricity of Djibouti).

The wind power plant, Djibouti's first utility-scale wind park, is built and owned by Red Sea Power Limited SAS, a simplified limited liability company incorporated under the laws of the Republic of Djibouti, registered within the Commercial Registry of Djibouti Number 15366/B/SAS, a Djiboutian registered company owned by the Africa Finance Corporation (51%), the Great Horn Investment Holdings (10%)¹, FMO Entrepreneurial Development Bank (19.5%) and Climate Investor One (CIO) (19.5%).

Djibouti imports over 70% of its energy from Ethiopia, dominated by hydropower plants, while the local energy generation solely comes from diesel-fueled power capacities. The project will thus substitute grid electricity by renewable energy and cut down GHG emissions from baseline fossil fuel intensive grid mix. Apart from emission reductions, the benefits of the project comprise, among others, improvement of energy self-sufficiency of the country, as well as creation of local employment.

It responds to the objectives of Djibouti as stated in its INDCs committing the country to increase the share of renewables in electricity production to 100%, as echoed in the National Energy Policy of 2013 in alignment with Vision 2035.

The project's 59.8MW will generate approximately 242 GWh per year and 988,499 tCO₂e of emission reductions over the ten years crediting period i.e. 98,849 tCO₂e of emission reductions per year.

Implementation timeline:

February 2018	Feasibility studies
August 2018	ESIA
May 19 th 2019	Power Purchase Agreement with Electricity de Djibouti
January 2020	Start of construction works
June 2022	Commissioning (expected)

¹ Post Construction AFC will sell 10% of its share in the project to GHIH

Prior to the implementation of the project, the existing scenario consisted in power generation and imports from Ethiopia existing grid mix, mainly composed of hydroelectric sources (78% of the total installed capacity and power generation²).

1.2 Sectoral Scope and Project Type

The sectoral scope is Scope 1 – Energy Industries (renewable - / non-renewable sources)

ACM-0002: Grid-connected electricity generation from renewable sources version 20.

The project is a renewable energy type and is not a grouped project.

1.3 Project Eligibility

The project is eligible under the scope of the VCS program version 4.1 because:

- It results in CO2 emission reduction, one of the six Kyoto protocol greenhouse gases
- It is supported by methodologies approved under a VCS approved GHG program
- The project activity, grid-connected electricity generation using (large scale) wind power in a Least Developed Country, is not excluded as per para 2.1.1 of the VCS Standard v4.1.

1.4 Project Design

The project has been designed to include a single installation of an activity.

Eligibility Criteria

N/A as not a grouped project

1.5 Project Proponent

Organization name	Red Sea Power Limited SAS
Contact person	Frank Maze
Title	Chief Executive Officer
Address	Moulk Centre Avenue Gandhi a Gabode 3 Republic of Djibouti
Telephone	+253 321 350 233

² <https://www.edd.dj/edd.html>

Email	f.maze@redseapower.dj
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1.6 Other Entities Involved in the Project

Organization name	AERA GROUP SAS
Role in the project	Carbon consultant
Contact person	Alexandre Dunod
Title	COO
Address	28 cours Albert 1er, 75008 Paris, France
Telephone	+33 1 42 18 02 02
Email	a.dunod@aera-group.fr

1.7 Ownership

The legal right to control and operate the project activity belongs to Red Sea Power Limited SAS, as per PPA dated 19th May 2019.

Aera Group has gained ownership through an Emissions Reduction Sale and Purchase Agreement on 14th December 2021 with Red Sea Power Limited SAS.

1.8 Project Start Date

The project start date is July 1st 2022, as the expected date of commissioning.

1.9 Project Crediting Period

From 01st of July 2022 until 30th June 2032. 10 years not renewable.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

The estimated annual GHG emission reductions/removals of the project are:

- ☐ <20,000 tCO₂e/year
- ☒ 20,000 – 100,000 tCO₂e/year
- ☐ 100,001 – 1,000,000 tCO₂e/year
- ☐ >1,000,000 tCO₂e/year

Project Scale	
Project	
Large project	X

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
From July 1 st 2022	38 930
2023	79 202
2024	81 953
2025	84 842
2026	87 874
2027	96 621
2028	103 814
Until June 30 st 2019	103 814
Total estimated ERs	988 499
Total number of crediting years	10
Average annual ERs	98 849

1.11 Description of the Project Activity

The proposed project consists of setting-up 17 G132-3.465MW T84 (capacity of 3.465 MW each). Generated electricity will be fed via aerial collector lines (i.e. cables) to a substation on the Project site. An overhead transmission line will connect the Project substation (located adjacent to the RN9 at approximately lat: 11.531628 long: 42.496840) to the 230 kV circuits located in the proposed EDD Ghoubet transformer/ substation, adjacent to the RN9 national road, 3.5 km from the Project site.

This clean energy supply thus implies a substantial power capacity increase for the country, whose totally dependent of the Ethiopian grid.

The main elements of the wind power plant are 17 G132-3.465MW T84 wind turbines of 3.465 MW each; whose specifications are detailed below:

Make	Siemens Gamesa
Model	G132
Power	3.465 MW
Cut-in wind speed [m/s]	3m/s

Cut-out wind speed [m/s]	25m/s																																																														
Rated wind speed [m/s]	14m/s																																																														
Rotor diameter [m]	132m																																																														
Converter voltage [V]	690 V at the output of the wind turbine generator converter 33 kV at the output of the wind turbine transformer																																																														
Hub height [m]	84m																																																														
Lifetime	20 years																																																														
Geo location	<table><tr><th colspan="3">COORDINATES OF WIND TURBINES</th></tr><tr><th colspan="3">COORDINATE SYSTEM UTM WGS84 ZONE 38 N</th></tr><tr><th>WTG</th><th>COORDINATE X</th><th>COORDINATE Y</th></tr><tr><td>WT 01</td><td>228988</td><td>1275041</td></tr><tr><td>WT 02</td><td>228839</td><td>1274670</td></tr><tr><td>WT 03</td><td>228686</td><td>1274301</td></tr><tr><td>WT 04</td><td>228419</td><td>1275355</td></tr><tr><td>WT 05</td><td>228074</td><td>1274792</td></tr><tr><td>WT 06</td><td>228169</td><td>1275919</td></tr><tr><td>WT 07</td><td>227810</td><td>1275366</td></tr><tr><td>WT 08</td><td>227548</td><td>1275947</td></tr><tr><td>WT 09</td><td>227178</td><td>1275554</td></tr><tr><td>WT 10</td><td>227104</td><td>1276351</td></tr><tr><td>WT 11</td><td>226876</td><td>1275989</td></tr><tr><td>WT 12</td><td>226329</td><td>1276881</td></tr><tr><td>WT 13</td><td>226453</td><td>1276425</td></tr><tr><td>WT 14</td><td>226079</td><td>1275968</td></tr><tr><td>WT 15</td><td>225738</td><td>1276849</td></tr><tr><td>WT 16</td><td>225693</td><td>1276439</td></tr><tr><td>WT 17</td><td>225571</td><td>1277291</td></tr></table>			COORDINATES OF WIND TURBINES			COORDINATE SYSTEM UTM WGS84 ZONE 38 N			WTG	COORDINATE X	COORDINATE Y	WT 01	228988	1275041	WT 02	228839	1274670	WT 03	228686	1274301	WT 04	228419	1275355	WT 05	228074	1274792	WT 06	228169	1275919	WT 07	227810	1275366	WT 08	227548	1275947	WT 09	227178	1275554	WT 10	227104	1276351	WT 11	226876	1275989	WT 12	226329	1276881	WT 13	226453	1276425	WT 14	226079	1275968	WT 15	225738	1276849	WT 16	225693	1276439	WT 17	225571	1277291
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Associated 2*60MVA transformers, before the metering equipment composed of 2 class 0.2S bi-directional meters (main and check meters) for import/export located at Goubet 230kV sub-station Delivery Point for EDD, as well as 2 class 0.2S bi-directional meter within the wind farm's premises for control purposes.

In accordance with the provisions of the Law No. 88/AN/15/7th Regulating the Activities of Independent Electricity Producers dated 01st of July 2015, Red Sea Power Ltd has been granted a Generation License by the Ministère de l'Energie et des Ressources Naturelles (MERN).

1.12 Project Location

The project is located between Lake Assal and Lake Goubet (1 km West of the lake) in the Arta region of the Republic of Djibouti.

Its central geo coordinates are approximately 11.5300000, 42.5000000 (38N coordinate system) or 11°31'48" N – 42°30'0" E.



Figure 1 : Site location of the project

1.13 Conditions Prior to Project Initiation

Djibouti's electricity is primarily generated by conventional thermal power stations in the country. However, with no proven reserves of oil or natural gas and no known coal reserves, electricity being produced is dependent on imported oil and therefore susceptible to price fluctuations. Djibouti also imports hydroelectric energy from Ethiopia via a high-voltage interconnection that has been in operation since 2012. The interconnection provides a cost-effective supply of electricity throughout the year. However, power imports are not guaranteed because Ethiopia regularly curtails supply, particularly during its dry season (September to February). During the rainy season (June to August), failures of the Ethiopian transport network often lead to unplanned interruptions. With national energy demand increasing between 3-5% annually, the country needs to develop new, more sustainable methods of energy production.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

To date, there are no regulations and policies preventing the implementation of the project activity, which complies with all and any relevant local, regional and national laws, statutes and regulatory frameworks:

Décret n° 2009-0218/PR/MERN Establishing the national Energy. This decree establishes the National Energy Commission, whose mission is to ensure the coordination of energy projects, and more generally to undertake studies of all the measures contributing to a better coordination of the country's energy development.

This Commission is responsible for intervening in the strategic areas of energy development in the Republic of Djibouti including studies, prospecting, research, exploration, exploitation and commercial.

Such a plant, like other renewable energy power plants planned will partly solve the problem of energy and achieving the objectives of a 100% electricity production from renewable energies goal by 2035.

In Djibouti, the procedure for evaluating impact assessments is governed by Djiboutian Government Law n° 51/AN/09/6ème L bearing the code of the environment establishing the basic rules and fundamental principles of national policy in the field of environmental protection and management.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project doesn't seek registration nor is registered under another GHG program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected under a GHG program.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

Does the project reduce GHG emissions from activities that are included in an emissions trading program or any other mechanism that includes GHG allowance trading?

☐ Yes

☒ No

If yes, provide the name of the emissions trading program or other mechanism that allows GHG allowance trading.

The project activity is not included in any emissions trading program or any other mechanism that includes GHG allowance trading.

1.16.2 Other Forms of Environmental Credit

Has the project sought or received another form of GHG-related credit, including renewable energy certificates?

☐ Yes☒ No

The project has not sought or received another form of GHG-related environmental credits including renewable energy certificates.

1.17 Sustainable Development Contributions

Besides emission reductions impacts and fuel cost savings, economic and social benefits of this \$122 million dollar wind power project to the local population include around 300 jobs created during construction, of which 10 permanent positions (local technicians, finance and administration) for the operation and maintenance of the plant.

Sustainable development, supported by the social, economic and environmental pillars, is the foundation of the National Energy Policy. In practice, (renewable) energy generation contributes to the improvement of the lives and livelihoods of Djiboutian's people, besides the nationally stated goal of a 100% electricity production from renewable energies by 2035.

Energy also supports the industrialisation of Djibouti and promotes local content and local participation to satisfy the Policy's economic requirements. With regards to environmental criteria, the use of (renewable) energy resources and energy consumption ensures that the environment and its multitude of services remain intact.

1.18 Additional Information Relevant to the Project

Leakage Management

The project does not generate any leakage.

Commercially Sensitive Information

There is no information deemed as commercially sensitive for this project activity.

Further Information

The project participants obtained all necessary clearances and licences; hence no legislative, economic, sectoral, social, environmental, geographic, site-specific risks are anticipated which may have impact on the eligibility of the project activity and the net GHG emission reductions.

2 SAFEGUARDS

2.1 No Net Harm

According to the Environmental and Social Impact Assessment (ESIA), the project presents a total of 18 impacts, and among those only 3 were assessed as being of either moderate or major significance: Communicable and Non-communicable diseases (moderate), Community safety and security (moderate) and Landscape and visual (major). The mitigation measures taken to address those impacts comprise the recruitment of a Community Liaison Officer, a worker EHS induction programme and the establishment of a site clinic to provide workers with medical treatment. Regarding the impacts on local landscape, the mitigation includes the limitation of land clearance restriction of site lighting and maintenance of tidy and contained site compounds. More information on the total listed impacts is available on section 2.3 matrix, while more details on them, mitigation measures but also positive impacts assessed are developed in the ESIA report, volume I and II.

2.2 Local Stakeholder Consultation

The following meetings were held with stakeholder, during the ESIA process from February to December 2018.

Stakeholder Engagement Meetings Held

Date	Stakeholder	Participants		
		Male	Female	Total
Government meetings				
12 Dec 2018	Electricité de Djibouti (EDD)	1		1
17-19 Feb 2018				
3 May 2018				
11 Feb 2018	Ministry of Agriculture	1		1
26 Feb 2018	Ministry of Habitat, Urban Planning, Environment and Town Planning (MHUE)	1		1
2 May 2018	Environment and Sustainable Development Directorate (part of MHUE)	2		2
	Total	5		5
Local level meetings				
8 Feb 2018	Arta Prefecture, meeting with Prefect	1		1
10 Feb 2018	Cité Moumina, Focus Group discussion with customary authorities	2		2
10 Feb 2018	Karta authority	1		1
11 Feb 2018	Tadjourah Prefecture, meeting with Prefect	1		1
14 Feb 2018	Public consultation in Lac Assal village. Attendees included: Sub-prefect and Village Chief of Lac Assal, members of local associations (including the Women's and Youth associations)	11	2	13
17 Feb 2018	Meeting with Okal, customary authority	1		1
17 Feb 2018	Public consultation in Cité Moumina community. Attendees included customary authorities and village elders.	6		6
19 Feb 2018	Focus Group discussion, Cité Moumina / Lac Assal Women's Association		2	2
18-19 Feb 2018	Individual interviews with Key informants (such as livestock breeders, fishermen, manager of the tourist resort) during the social baseline engagement.	7		7
15-19 Feb 2018	Household surveys were completed with 40 households in the Project area during the social baseline engagement.			40
3 May 2018	Public consultation in Lac Assal's Community Building. Attendees included: Sub-prefect and Chief of Lac Assal, Lac Assal Women's Association, Okal General, Imam of Cité Moumina Mosque, Makaban (customary authorities representing the Debné tribes) and community members.	14	2	
	Total	30	6	74
Grand Total		35	6	79

During the public consultation meetings discussions with stakeholders included:

- presentations of the Project; and
- discussions on the potential impacts of the Project, on the proposed solutions to minimise negative impacts and maximise positive benefits, and on the overall expectations regarding the project.

Stakeholder engagement will continue following completion of the ESIA process. A project SEP will be developed and will include the following key elements:

- Pre-Construction Engagement: The project will ensure that there are sufficient community relations resources on the ground to manage grievances and undertake engagement regarding timeframes for construction activities. A series of meetings will be held in impacted areas, including along the transportation route with affected people.
- Construction Engagement: During this phase, the community relations team will respond to grievances and continue to hold meetings with stakeholders as required to address issues of concern. In addition, the team will monitor the implementation of the mitigation measures. The number of grievances and meetings is likely to be greatest during the construction period due to the nature and extent of predicted impacts.
- Operations Engagement: Ongoing engagement will be undertaken as required to manage community concerns and issues as well as to address grievances. Meeting frequency and grievances are expected to reduce over time as stakeholders adapt to the presence of the windfarm.

Grievance mechanism:

An effective grievance mechanism allows stakeholders to lodge complaints and/or concerns at no cost, without retribution and with the assurance of a timely response. The process will include the following steps:

- identification;
- review and record the grievance;
- acknowledgement;
- develop a response;
- communicate response and establish agreement; and
- close-out process.

A detailed description of the grievance mechanism required for the Project is provided in the SEP.

Stakeholders can submit grievances or comments regarding the Project to the Community Liaison Officer or through local community leaders. Prior to construction, an office will be established at the site to improve accessibility of the grievance process.

2.3 Environmental Impact

Significance	Significance Context
Negligible	A resource/receptor (including people) will essentially not be affected in any way by a particular activity or the predicted effect is deemed to be 'imperceptible' or is indistinguishable from natural background variations.
Minor	A resource/receptor will experience a noticeable effect, but the impact magnitude is sufficiently small (with or without mitigation) and/or the resource/receptor is of low sensitivity/ vulnerability/ importance. In either case, the magnitude should be well within applicable standards.
Moderate	Has an impact magnitude that is within applicable standards, but falls somewhere in the range from a threshold below which the impact is minor, up to a level that might be just short of breaching a legal limit.
Major	An accepted limit or standard may be exceeded, or large magnitude impacts occur to highly valued/sensitive resource/receptors.

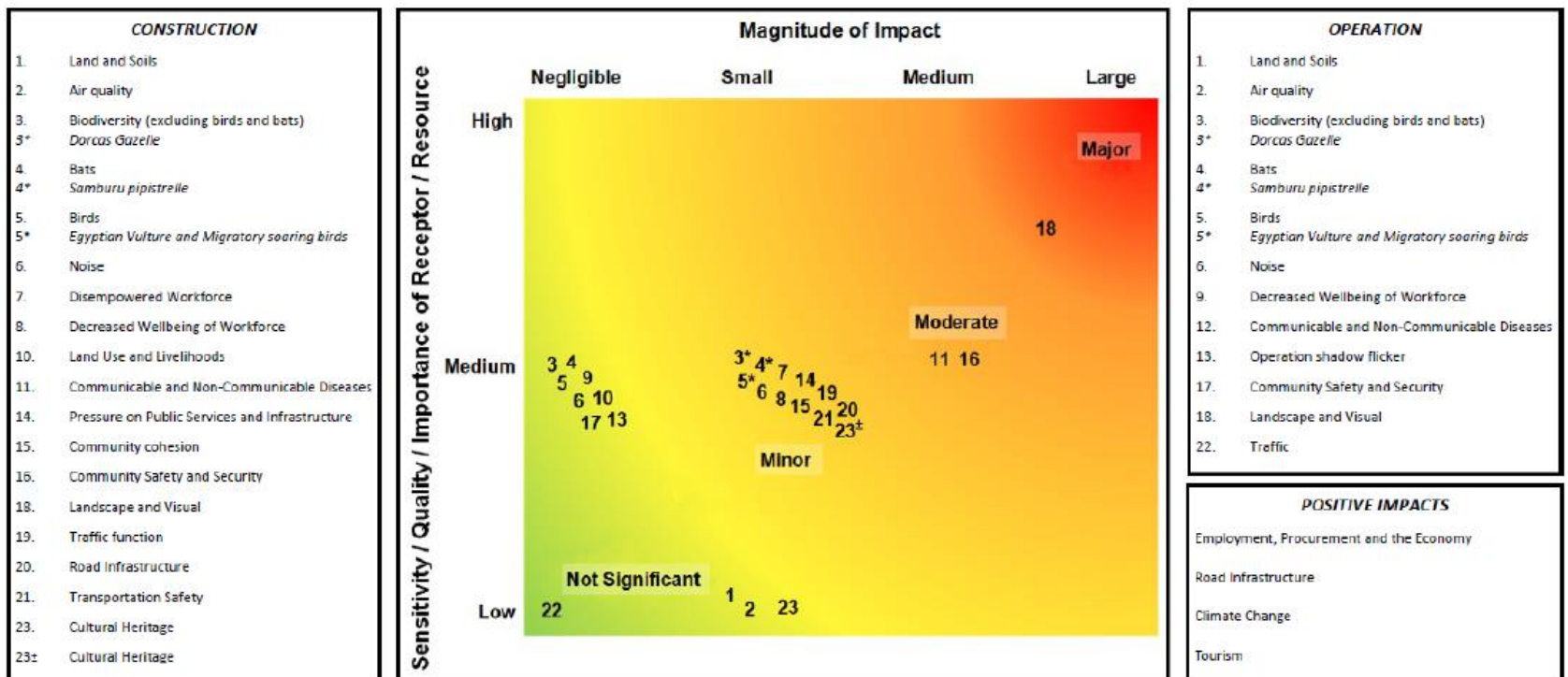


Figure 2 : ESIA assessed impacts matrix

2.4 Public Comments

The following table provides a summary of issues raised and how they have been addressed in the ESIA :

Subject	Issue	Project Response	ESIA Reference
Land Use and Land Take	- The Project footprint does not constitute a major issue for the community. The affected land is unsuitable for any productive activity, and the Project is thus not expected to hinder any agricultural or pastoral projects of the residents. - Some concerns were raised about the project actually materialising: reportedly projects have been planned in the past with no follow through.	- Access will be temporarily restricted in small areas of the site during construction. The Project will engage with communities beforehand to minimise the impact and agree on alternative accessible routes. - The Project will employ a Community Liaison Officer who will communicate all Project developments to the communities so that they are informed of Project development progress.	<i>Section 7.12 (Land Use and Livelihoods)</i>
Employment and the Economy	- Community members have high expectations in terms of access to jobs, both during the construction phase of the project and as a result of the electrification of the area. - Some members expressed concerns that outsiders will take the jobs, due to the lack of schooling and professional training of the local population – and this will reportedly be unacceptable for them.	- The Project will implement a Human Resources Policy and Management Plan setting out recruitment procedures, training requirements and targets around the hiring of local workers. - Local procurement will be prioritised where possible. - The Project will provide information on the specific jobs available and specific skills required, at least two months prior to the hiring period. This is to ensure opportunities are known in advance for unskilled and semi-skilled positions, which are more likely to be filled by local residents.	<i>Section 7.11 (Employment, Procurement and Economy)</i>
Demography and immigration	- In Lac Assal, potential demographic increase as a result of the project is seen as an opportunity. Some concerns were raised related to unplanned or chaotic urbanisation, and waste management. - In Cité Moumina, the community members maintained that the presence of workers or jobseekers from other areas will be a concern for them, and thus the village elders should be updated continuously to manage any potential influx of people.	- The Project will aim to detract work-seeking immigration to the Project area through the clear communication of Project labour needs and clear Project policies and procedures for recruitment. - The Project will implement an Influx Monitoring Programme and an Influx Management Plan (if influx is identified as a growing concern during monitoring programme) - The Project will implement a Grievance Mechanism that is easily accessible to local communities	<i>Section 7.13.7 (Community Health, Safety and Security)</i>

Environment	The community members were given the opportunity to ask questions regarding windfarms - noise pollution was the theme that raised most concerns.	- Noise emissions from construction will be temporary and limited to the duration of the equipment operations at the work sites. - It is planned that turbines will be sited at least 500 m from any dwellings to mitigate noise impacts. - The Project will commit to the IFC EHS General Guidelines on noise impacts: 45 dB(A) limit during operation at residential receptors.	Section 7.10 (Noise)
Health and wellbeing	- No particular concerns related to health and safety during the construction phase, especially if local people were to be in charge of site safety. - Questions related to the safety of the area once the turbines have been installed (lightening and fears that the rotor blades may detach and fall).	- The Project will engage with the local community through the Community Liaison Officer about the risks and safety features of the wind turbines; turbine failure resulting in 'turbine throw' is considered unlikely. - Set-back distances (≥500 m) between turbines and residential areas will be incorporated into the design. - The Project will engage with the local community on what to do in the event of an emergency scenario such as turbine failure.	Section 7.13 (Community Health, Safety & Security) Section 8 (Unplanned Events)
Cultural heritage	There were concerns that the cemeteries located just outside the project perimeter will be damaged by the Project.	The cemeteries are located outwith the Project site therefore direct impacts to them will be avoided. A suitable buffer has also been included in the turbine layout design.	Section 6.10 (Cultural Heritage)
Engagement during the Project	Recommended means to set up a good communication and information sharing system between the project and the local communities.	- During the public consultation meetings, the EDD representative noted that there are plans for weekly meetings with the community. This plan was welcomed by the community. - The Project will employ a Community Liaison Officer who will communicate all Project developments to the communities. - A community grievance mechanism has been communicated and participants welcomed this.	Annex G Stakeholder Engagement Plan

2.5 AFOLU-Specific Safeguards

Not an AFOLU project

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

The approved baseline and monitoring methodology selected for the proposed project activity is:

ACM0002 “Grid connected electricity generation from renewable sources” version 20.

<https://cdm.unfccc.int/methodologies/DB/XP2LKUSA61DKUQC0PIWPGWDN8ED5PG>

The tools used are:

 [Tool for the demonstration and assessment of additionality](#) (232 KB)

 [Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation](#) (559 KB)

 [Tool to calculate the emission factor for an electricity system](#) (1202 KB)

3.2 Applicability of Methodology

The choice of the ACM0002 methodology is accurate since the proposed project activity respects all the applicability conditions required.

ACM0002 version 20 applicability conditions	Project activity applicability
This methodology is applicable to grid-connected renewable energy power generation project activities that: a) Install a Greenfield power plant; b) Involve a capacity addition to (an) existing plant(s); c) Involve a retrofit of (an) existing operating plants/units; d) Involve a rehabilitation of (an) existing plant(s)/unit(s); or e) Involve a replacement of (an) existing plant(s)/unit(s).	The project activity is a greenfield wind power plant substituting electricity produced on the grid by renewable energy.
The project activity may include renewable energy power plant/unit of one of the following types: - hydro power plant/unit (with or without reservoir), - wind power plant/unit, - geothermal power plant/unit, - solar plant/unit, - wave power plant/unit or - tidal power plant/unit;	The project activity is the construction and operation of a wind power plant and hence the methodology is applicable.
In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a	The project activity does not involve any capacity

ACM0002 version 20 applicability conditions	Project activity applicability
minimum historical reference period of five years, used for the calculation of baseline emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity.	additions, retrofits, rehabilitations or replacements.
In case of hydro power plants, one of the following conditions shall apply: a) The project activity is implemented in existing single or multiple reservoirs, with no change in the volume of any of the reservoirs; or b) The project activity is implemented in existing single or multiple reservoirs, where the volume of the reservoir(s) is increased and the power density calculated using equation (7) of the methodology ACM0002, is greater than 4 W/m²; or c) The project activity results in new single or multiple reservoirs and the power density, calculated using equation (7) of the methodology ACM0002, is greater than 4 W/m²; or d) The project activity is an integrated hydro power project involving multiple reservoirs, where the power density for any of the reservoirs, calculated using equation (7) of the methodology ACM0002, is lower than or equal to 4 W/m², all of the following conditions shall apply: - The power density calculated using the total installed capacity of the integrated project, as per equation (8) of the methodology ACM0002, is greater than 4 W/m²; - Water flow between reservoirs is not used by any other hydropower unit which is not a part of the project activity; - Installed capacity of the power plant(s) with power density lower than or equal to 4 W/m² shall be: a.) Lower than or equal to 15 MW; and b.) Less than 10 per cent of the total installed capacity of integrated hydro power project.	Not applicable as the proposed project activity involves a wind power plant.
In the case of integrated hydro power projects, project proponent shall: - Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or - Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum five years prior to implementation of CDM project activity.	Not applicable as the proposed project activity involves a wind power plant.
The methodology is not applicable to: - Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; - Biomass fired power plants/units.	The proposed project activity neither involves - switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may

ACM0002 version 20 applicability conditions	Project activity applicability
	be the continued use of fossil fuels at the site, nor - biomass fired power plants/units.
In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project activity does not involve capacity additions, retrofits, rehabilitations or replacements.
In addition, the applicability conditions included in the tools referred to above apply.	Applicability conditions of the applied tool are justified.

Table 2: Compliance of the project activity regarding ACM0002 applicability conditions.

The project activity meets the following applicability conditions of TOOL07 “Tool to calculate the emission factor for an electricity system” (Version 7.0).

No	Applicability condition	Applicability to this project activity
1	This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	As part of ACM0002, “operating margin” (OM), “build margin” (BM) and “combined margin” (CM) need to be estimated to calculate baseline emissions of the project activity that substitutes electricity in the Djiboutian grid. Hence the tool is applicable.
2	Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off - grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If option IIa is chosen, the conditions specified in “Appendix 2: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity	The emission factor for the project electricity system is calculated for grid power plants and off-grid power plants. Option IIb is applied, i.e. the tool is applicable.

No	Applicability condition	Applicability to this project activity
	generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
3	In case of CDM projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Since the project electricity system is not located partially or totally in an Annex I country - it is located in the Republic of Djibouti - the tool is applicable.
4	Under this tool, the value applied to the CO ₂ emission factor of biofuels is zero.	There are no biofuels used in the project activity, i.e. the tool is applicable.

Table 3: Compliance of the project activity project activity regarding applicability conditions of “Tool to calculate the emission factor for an electricity system”.

The project activity also meets the following applicability conditions of TOOL05: “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation” (Version 03.0).

No	Applicability condition	Applicability to this project activity
1	This tool can be referred to in methodologies to provide procedures to monitor amount of electricity generated in the project scenario, only if one out of the following three project scenarios applies to the recipient of the electricity generated: (a) Scenario I: Electricity is supplied to the grid; (b) Scenario II: Electricity is supplied to consumers/electricity consuming facilities; or (c) Scenario III: Electricity is supplied to the grid and consumers/electricity consuming facilities.	As per ACM0002, V.20, “EG _{facility,y} should be determined as per “TOOL05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation.” As per para. 22 of ACM0002, one possible project scenario is the installation of a Greenfield power plant delivering electricity to the grid, i.e. at least scenario I applies and the tool can be referred to.
2	This tool is not applicable in cases where captive renewable power generation technologies are installed to provide electricity in the project activity, in the baseline scenario or to sources of leakage. The tool only accounts for CO ₂ emissions.	No captive renewable power generation technology is installed.

Table 4: Compliance of the project activity project activity regarding applicability conditions of tool “Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation”

Other tools mentioned in the methodology are not applicable to this project activity.

3.3 Project Boundary

Source		Gas	Included?	Justification/Explanation
Baseline	CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
		Other	No	Minor emission source
Project	For geothermal power plants, (fugitive) emissions of CH ₄ and CO ₂ from non-condensable gases contained in geothermal steam	CO ₂	No	Not applicable (Only for geothermal)
		CH ₄	No	Not applicable (Only for geothermal)
		N ₂ O	No	Not applicable
		Other	No	Not applicable
	For binary geothermal power plants, fugitive emissions of hydrocarbons such as n-butane and isopentane (working fluid) contained in the heat exchangers	Low GWP hydro-carbon/ refrigerant	No	Not applicable (Only for geothermal)
		CO ₂	No	Not applicable (Only for solar thermal power plants and geothermal power plants)
		CH ₄	No	Not applicable
		N ₂ O	No	Not applicable
	CO ₂ emissions from combustion of fossil fuels for electricity generation in solar thermal power plants and geothermal power plants	CO ₂	No	Not applicable (Only for solar thermal power plants and geothermal power plants)
		CH ₄	No	Not applicable
		N ₂ O	No	Not applicable
		Other	No	Not applicable
	For hydro power plants, emissions of CH ₄ from the reservoir.	CO ₂	No	Not applicable (Only for hydro)
		CH ₄	No	Not applicable
		N ₂ O	No	Not applicable
		Other	No	Not applicable

According to ACM0002 methodology, the spatial extent of the project boundary includes the project power plant and all power plants connected physically to the electricity system that the VCS project power plant is connected to. The project boundary is therefore determined as:

- the project activity site, where the electricity is being produced,
- the grid that the power plant is connected to.

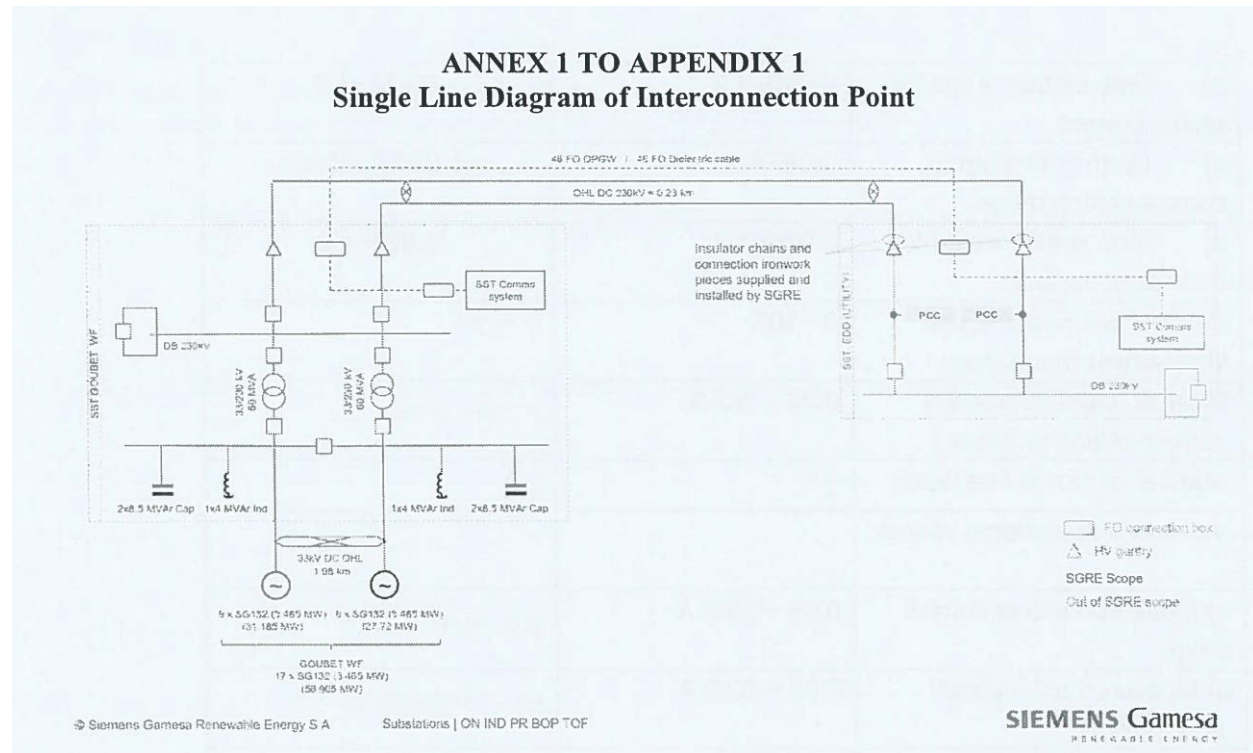


Figure 3 : Single line diagram

3.4 Baseline Scenario

According to ACM0002 Version 20.0 and since the project activity is the installation of a new grid-connected renewable power plant (Greenfield) the baseline scenario is the following:

“Electricity delivered to the grid by the project activity would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in the “Tool to calculate the emission factor for an electricity system.”

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as described in section 4.1

3.5 Additionality

According to ACM0002 Version 20, the tool 01, “Tool for the demonstration and assessment of additionality” version 07.0.0 applies for additionality demonstration, as well as the Tool 23 Additionality of first of its kind project activities version 03.0.

The step 0 from the methodology procedure of the tool for the demonstration and assessment of additionality is to demonstrate whether the proposed project activity is the first of its kind.

The tool 23 Additionality of first of its kind project activities states that a proposed project activity is the first of its kind in the applicable geographic area (host country) if:

Applicable area	Entire host country Djibouti (a)
Measure	Switch of technology with change of energy source (b) power generation based on renewable energy
Output	Goods/services produced by the project activity Electricity
Different technologies	Size of installation (power capacity) Large
Crediting period	The project participants selected a crediting period for the project activity that is “a maximum of 10 years with no option of renewal” (c)
Evidence	As of project’s start date (01/07/2022), no other large-scale wind power plant (>15 MW) was in commercial operation in Djibouti.

The project is thus considered additional.

3.6 Methodology Deviations

No methodology deviations were used.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

Baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity. The methodology assumes that all project electricity generation above baseline levels would have been generated by existing grid-connected power plants and the addition of new grid-connected power plants. The baseline emissions are to be calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO ₂ /MWh)

Calculation of $EG_{PJ,y}$

Since the project activity consists of the installation of new grid-connected renewable power plant at a site where no renewable power plant was operated prior to the implementation of the project activity, the project activity is the installation of a greenfield renewable energy power plant, so that:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the project activity in year y (MWh/yr)
$EG_{facility,y}$	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Calculation of $EF_{grid,CM,y}$

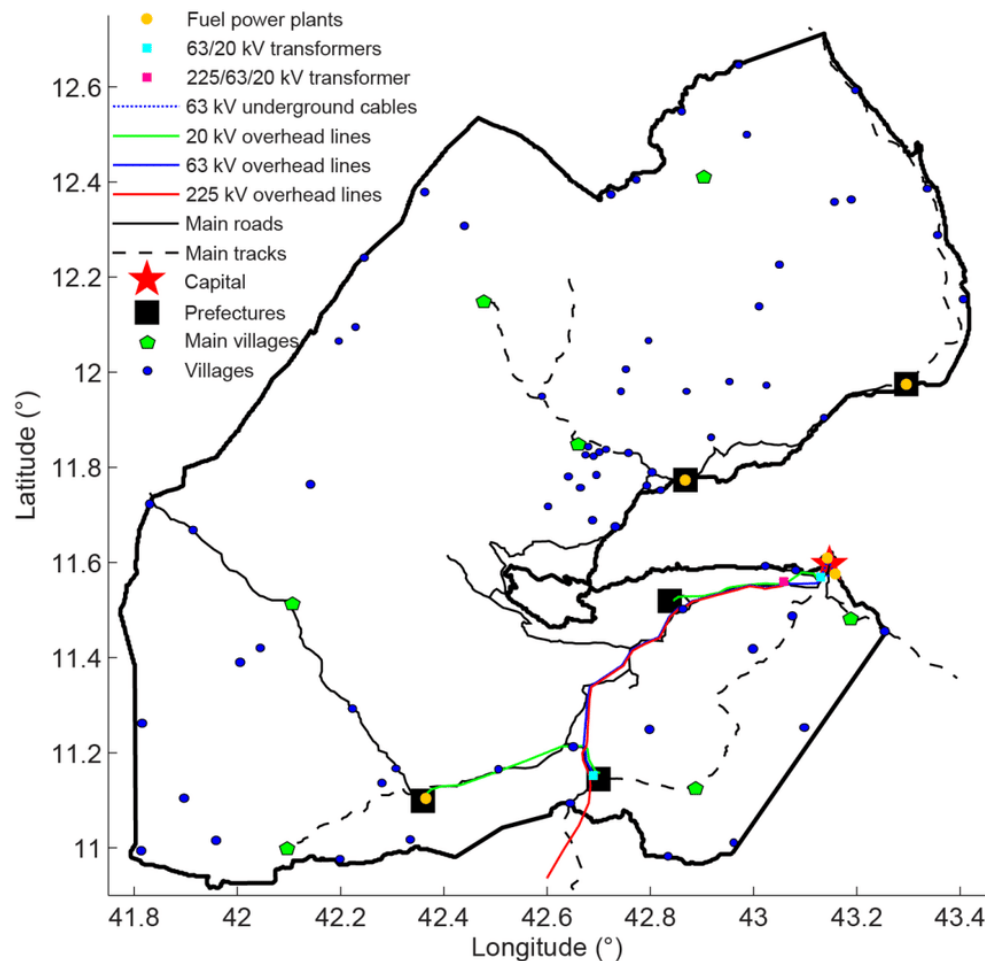
The grid emission factor ($EF_{grid,CM,y}$) is calculated ex-ante as per the “Tool to calculate the emission factor for an electricity-system” (Version 7.0). The emission factor is not monitored during the crediting period of the project activity but shall be updated at the renewal of the crediting period of the project activity.

This methodological tool determines the CO₂ emission factor for the displacement of electricity generated by power plants in an electricity system, by calculating the “combined margin” emission factor (CM) of the electricity system. The CM is the result of a weighted average of two emission factors pertaining to the electricity system: the “operating margin” (OM) and the “build margin” (BM). The operating margin is the emission factor that refers to the group of existing power plants whose current electricity generation would be affected by the project activity. The build margin is the emission factor that refers to the group of prospective power plants whose construction and future operation would be affected by the project activity.

The tool indicates six steps for the calculation of the combined margin (CM) emission factor:

STEP 1. Identify the relevant electricity systems.

The following map shows the geographical boundary of the Djiboutian grid.



As of 2021 (latest data available before start date), the DNA of the host country has not published a delineation of the project electricity system and connected electricity systems. Thus, the project participant defines the project electricity system and interconnected systems as per Option 2 based on EDD delineation (national utility responsible for electricity dispatching).

Therefore, the relevant electric power system is the national grid of Djibouti, including all power plants physically connected through transmission and distribution lines to the project activity. It is managed by EDD, which annually aggregated power production and imports summaries have been obtained and relied on for the establishment of the grid emission factor calculation.

STEP 2. Choose whether to include off-grid power plants in the project electricity system (optional).

The following option is selected to calculate the operating margin and build margin emission factor:

- Option II: Both grid power plants and off-grid power plants are included in the calculation.

STEP 3: Select a method to determine the operating margin (OM)

The calculation of the operating margin emission factor ($EF_{grid,OM,y}$) is based on the following method, which is described under Step 4:

(b) Simple adjusted OM;

The Simple OM method (a) can be used when low-cost/must run resources constitute less than 50% of the total amount of the power generation on the grid, in average of the five most recent years. Between 2017 and 2021 the low-cost/must run share represented more than 50%. Therefore, method (a) Simple OM is inapplicable.

The simple adjusted OM method is applied since:

- LCMR share is greater than 50% in recent 5 years
- Average load by LCMR is greater than average LASL over three years
- Hourly loads of the grid in MW are unavailable and $LASL > 1/3 HASL$

Table 6: Net power generation and share of low cost/must run power plants

	5 years of historical data				
Year	2017	2018	2019	2020	2021
Net production [MWh]	557 312	569 402	625 784	646 695	652 193
Total share of low-cost/must-run [%]	91,2%	92,4%	79,0%	82,2%	83,1%

The simple adjusted OM emission factor ($EF_{grid,OM-adj,y}$) is thus applied. It is a variation of the simple OM, where the power plants/units (including imports) are separated in low-cost/must-run power sources (k) and other power sources (m).

STEP 4. Calculate the operating margin emission factor according to the selected method.

According to the tool, the simple adjusted OM emission factor ($EF_{grid,OM-adj,y}$) is calculated based on the net electricity generation of each power unit and an emission factor for each power unit, as follows:

$$EF_{grid,OM-adj,y} = (1 - \lambda_y) \times \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} + \lambda_y \times \frac{\sum_k EG_{k,y} \times EF_{EL,k,y}}{\sum_k EG_{k,y}}$$

Where:

- $EF_{grid,OM-adj,y}$ = Simple adjusted operating margin CO₂ emission factor in year y (tCO₂/MWh).
- λ_y = Factor expressing the percentage of time when low-cost/must-run power units are on the margin in year y
- $EG_{m,y}$ = Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh).
- $EG_{k,y}$ = Net quantity of electricity generated and delivered to the grid by power unit k in year y (MWh).
- $EF_{EL,m,y}$ = CO₂ emission factor of power unit m in year y (tCO₂/MWh)
- $EF_{EL,k,y}$ = CO₂ emission factor of power unit m in year y (tCO₂/MWh)
- m = All power units serving the grid in year y except low-cost/must-run power units.
- k = All low-cost/must run grid power units serving the grid in year y
- y = The relevant year as per the data vintage chosen in Step 3.

Under this option, $EF_{EL,m,y}$, $EF_{EL,k,y}$, $EG_{m,y}$ and $EG_{k,y}$ are determined using the same procedures as those for the parameters $EF_{EL,m,y}$ and $EG_{m,y}$ in Option A of the simple OM method. For grid power plants, $EG_{m,y}$ is determined as per the provisions in the monitoring tables of the tool, i.e. for the ex-ante option “once for each crediting period using the most recent three historical years for which data [from utility or government records or official publications] is available at the time of submission of the PDD to the DOE for validation.”

Since data on fuel consumption is not available, the emission factor ($EF_{EL,m,y}$) is determined as follows (option A2 of the tool):

$$EF_{EL,m,y} = \frac{EF_{CO2,m,i,y} \times 3.6}{\eta_{m,y}}$$

Where:

- $EF_{EL,m,y}$ = CO₂ emission factor of power unit m in year y (tCO₂/MWh)
- $EF_{CO2,m,i,y}$ = Average CO₂ emission factor of fuel type i used in power unit m in year y (t CO₂/GJ)
- $\eta_{m,y}$ = Average net energy conversion efficiency of power unit m in year y (ratio)
- m = All power units serving the grid in year y except low-cost/must-run power units
- y = The relevant year as per the data vintage chosen in Step 3.
- 3,6 = Conversion factor (GJ/MWh)

Off-grid power plants are included in the operating margin emission factor. The following default values can be used to determine $EG_{m,y}$ for the off-grid plants:

(a) The value of 10 per cent of the total electricity generation by grid power plants in the electricity system for the purpose of the operating margin determination;

Based on years 2019-2021, the calculated **Operating Margin is: 0.4085 tCO₂/MWh**.

STEP 5. Calculate the build margin (BM) emission factor.

No data to calculate the Build Margin is available because:

- There are no 5 most recent power units
- The Djibouti grid power units do not comprise at least 20 per cent of the system generation

STEP 6. Calculate the combined margin (CM) emission factor.

The combined margin emission factor is calculated following the simplified CM approach, because the grid is located in LDC/SIDs/URC.

$$EF_{grid,CM,y} = EF_{grid,OM,y}$$

Based on years 2019-2021, the **combined margin emission factor** and grid emission factor value used to calculate the emission reductions of the wind power plant project is **0.4085 tCO₂/MWh**

4.2 Project Emissions

Project emissions are calculated as follows:

$$PE_y = PE_{FF,y} + PE_{GP,y} + PE_{HP,y}$$

Where:

PE_y	=	Project emissions in year y (t CO ₂ e/yr)
$PE_{FF,y}$	=	Project emissions from fossil fuel consumption in year y (t CO ₂ /yr)
$PE_{GP,y}$	=	Project emissions from the operation of geothermal power plants due to the release of non-condensable gases in year y (t CO ₂ e/yr)
$PE_{HP,y}$	=	Project emissions from water reservoirs of hydro power plants in year y (t CO ₂ e/yr)

$PE_{FF,y}$, $PE_{GP,y}$ and $PE_{HP,y}$ are equal to 0 as the project is an installation of a wind power plant with no auxiliary fossil fuel consumption. In particular, ACM0002 §33 stipulates that for all renewable energy power generation project activities, emissions due to the use of fossil fuels for the backup generator can be neglected.

4.3 Leakage

No leakage emissions are considered. The emissions potentially arising due to activities such as power plant construction and upstream emissions from fossil fuel use (e.g. extraction, processing, transport etc.) are neglected.-

4.4 Net GHG Emission Reductions and Removals

According to the approved methodology ACM0002 ver.20.0, emission reductions are calculated as follows:

$$ER_y = BE_y - PE_y$$

Where:

ER_y = Emission reductions in year y (t CO₂e)

BE_y = Baseline emissions in year y (t CO₂)

PE_y = Project emissions in year y (t CO₂e)

Table 8: Calculation of emission reductions

	Value/Result	Source/reference
Total installed capacity	59.8 MW	Project documents
Net electricity delivered to the grid ($EG_{PJ,y}$)	242 GWh/yr	Project documents (nominal average) $[EG_{PJ,y} = EG_{facility,y}]$
Grid emission factor ($EF_{grid,CM,y}$)	0.4085 tCO ₂ e/MWh	Section 4.1
Baseline emissions (BE_y)	98,849 tCO ₂ e	Section 4.1 $BE_y = EG_{PJ,y} \cdot EF_{grid,CM,y}$
Project emissions (PE_y)	0 tCO ₂ e	Section 4.2
Emission reductions (ER_y)	98,849 tCO ₂ e	$ER_y = BE_y - PE_y$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
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01/07/2022 - 31/12/2022	38 930	-	-	38 930
2023	79 202	-	-	79 202
2024	81 953	-	-	81 953
2025	84 842	-	-	84 842
2026	87 874			87 874
2027	96 621			96 621
2028	103 814			103 814
2029	103 814	-	-	103 814
2030	103 814	-	-	103 814
2031	103 814	-	-	103 814
01/01/2032 - 30/06/2032	103 814	-	-	103 814
Total	988 499	-	-	988 499

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	EF _{grid,CM,y}
Data unit	tCO ₂ /MWh
Description	Combined margin CO2 emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” v7.0
Source of data	<i>Electricité De Djibouti grid data (Données Energétique EDD 2022.xlsx)</i>
Value applied	0,40854
Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” v7.0

Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	EF _{grid,OM,y}
Data unit	tCO ₂ /MWh
Description	Operating Margin CO ₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” v7.0
Source of data	<i>Electricité De Djibouti grid data (Données Energétique EDD 2022.xlsx)</i>
Value applied	0,40854
Justification of choice of data or description of measurement methods and procedures applied	As per the “Tool to calculate the emission factor for an electricity system” v7.0
Purpose of Data	Calculation of baseline emissions
Comments	-

5.2 Data and Parameters Monitored

Data / Parameter	EG _{facility,y}
Data unit	MWh/yr
Description	Provide a brief description of the data/parameter
Source of data	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y
Description of measurement methods and procedures to be applied	Electricity outputs will be electronically stored by the Technical/Engineering/ Maintenance Department under the Plant Manager’s authority
Frequency of monitoring/recording	Continuous measurement and at least monthly recording
Value applied	241,957
Monitoring equipment	<i>Bidirectional electricity meters</i>

QA/QC procedures to be applied	Calibration of the meters will be carried out as per manufacturer specifications. Annual inspection and testing of accuracy will be carried out by the project proponent and EDD as per PPA.
Purpose of data	Calculation of baseline emissions
Calculation method	Electronic recording
Comments	-

5.3 Monitoring Plan

The proposed project activity monitoring plan complies with the methodology ACM0002 - Consolidated baseline methodology for grid-connected electricity generation from renewable sources (Version 20.0), whereby it is stated that:

“All data collected as part of monitoring should be archived electronically and be kept at least for 2 years after the end of the last crediting period. 100% of the data should be monitored if not indicated otherwise in the tables of Section 6.1 of ACM0002 Ver. 17. All measurements should be conducted with calibrated measurement equipment according to relevant industry standards”.

Therefore, the quantity of net electricity generation supplied by the project plant to the grid will be reliably monitored through calibrated electricity meters and cross-checked with sales records as per internal monitoring practices.

As concerns the monitoring, Red Sea Power management has the overall responsibility for all VCS monitoring of the project, including:

- Develop, approve, execute, and improve the VCS Monitoring/Reporting Procedures;
- Organize in-house seminar to inform and train the company staff to the monitoring procedures;
- Ensure that instrumentations and devices are available and properly suited to efficiently perform the monitoring;
- Communicate and coordinate the monitoring work of all business units;
- Validate and electronically archive all monitoring data on a monthly basis throughout the crediting period (and conserve it at least for 2 further years);
- Calculate and report the emission reductions; and
- Coordinate the DOE work during the verification audit.

A VCS coordinator might be appointed with delegation of the above specific tasks of monitoring supervision. Recorded data is immediately collected and managed in user-friendly, detailed reports and tables to facilitate analysis.

Monitoring team and training

Data collection, consolidation and results analysis will be undertaken by a dedicated team adequately trained, well aware of VCS requirements. This team will not have any hierarchical relationships or dependence links with all entities involved to measure net electricity supplied to the grid and to assure the correct operation and maintenance of the measuring equipment. This independence shall guarantee the integrity of the work that will be done.

Emergency and trouble-shooting procedures

The immediate response capability will be achieved through the implementation of a system of supervision and control that will transfer real-time all information about the state of the equipment. The operator will provide a team of qualified stakeholders that can react in "real time".

